Arithmetic Sequences and Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Arithmetic sequences

Consider the sequence $7, 11, 15, 19, \ldots$

a State the common difference. $d = 4$

b Write a formula for the nth term. $a_n = 4n + 3$

c Find a_{20}. $a_{20} = 83$

(a) $d = 4$; (b) $a_n = 4n + 3$; (c) $a_{20} = 83$

2 [Foundational] • Target: Arithmetic sequences

In an arithmetic sequence, $a_3 = 10$ and $a_8 = 25$.

a Find the common difference and the first term. $5d = 15$ gives $d = 3$; $a_1 = 10 - 2(3) = 4$.

b Write a formula for a_n. $a_n = 4 + 3(n - 1) = 3n + 1$.

(a) $5d = 15$ gives $d = 3$; $a_1 = 10 - 2(3) = 4$.; (b) $a_n = 4 + 3(n - 1) = 3n + 1$.

3 [Foundational] • Target: Arithmetic series

Find the sum of the first 30 terms of the series $5 + 9 + 13 + \cdots$

$a_{30} = 5 + 29(4) = 121$, so $S_{30} = \frac{30(5 + 121)}{2} = 1890$.

4 [Foundational] • Target: Arithmetic series

Consider the multiples of 6 strictly between 100 and 500.

a How many are there? First 102, last 498: $n = \frac{498 - 102}{6} + 1 = 67$.

b Find their sum. $S = \frac{67(102 + 498)}{2} = 67 \times 300 = 20{,}100$.

(a) First 102, last 498: $n = \frac{498 - 102}{6} + 1 = 67$.; (b) $S = \frac{67(102 + 498)}{2} = 67 \times 300 = 20{,}100$.

5 [Foundational] • Target: Arithmetic series

A theatre has 18 seats in the first row, and each subsequent row has 3 more seats than the row before. How many seats are there in total in 25 rows?

$a_{25} = 18 + 24(3) = 90$, so $S_{25} = \frac{25(18 + 90)}{2} = 1350$ seats.

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6 [Foundational] • Target: Arithmetic means and sequence properties

Insert three arithmetic means between 5 and 25 (i.e., find b_1, b_2, b_3 so that $5, b_1, b_2, b_3, 25$ is arithmetic).

5 terms total, so $d = \frac{25-5}{4} = 5$. Means: $10, 15, 20$.

7 [Foundational] • Target: Arithmetic means and sequence properties

Given the arithmetic sequence with $a_1 = -8$ and $d = 3$, find the smallest n such that $a_n > 100$.

$a_n = -8 + 3(n-1) = 3n - 11 > 100$: $3n > 111$, $n > 37$: smallest integer $n = 38$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Arithmetic means and sequence properties

Prove that the sum of the first n positive odd integers $(1 + 3 + 5 + \cdots)$ equals n^2, using the arithmetic series formula.

The nth odd number is $a_n = 2n - 1$, with $a_1 = 1$, $d = 2$. $S_n = \frac{n(a_1 + a_n)}{2} = \frac{n(1 + 2n - 1)}{2} = \frac{n(2n)}{2} = n^2$.

9 [Intermediate] • Target: Applications

A person saves 100 riyals in week 1, and increases savings by 20 riyals each subsequent week. Find the total saved after 20 weeks.

$a_{20} = 100 + 19(20) = 480$. $S_{20} = \frac{20(100 + 480)}{2} = 5800$ riyals.

10 [Intermediate] • Target: Applications

A stack of logs has 20 logs in the bottom row, 19 in the next, decreasing by 1 each row, ending with 1 log in the top row. Find the total number of logs.

This is the sum $1 + 2 + \cdots + 20 = \frac{20(1 + 20)}{2} = 210$ logs.

11 [Intermediate] • Target: Visualizing an arithmetic sequence

The first 6 terms of the sequence $a_n = 3n - 1$ are plotted below. Use the graph to confirm the sequence is linear in n, and state the common difference from the graph.

The points $(1, 2), (2, 5), (3, 8), (4, 11), (5, 14), (6, 17)$ lie on a straight line, confirming the linear (arithmetic) pattern. The common difference $d = 3$ matches the constant rise between consecutive points.

12 [Intermediate] • Target: Arithmetic sequences in geometry

The interior angles of a pentagon form an arithmetic sequence. If the smallest angle is $80°$ and the angles sum to $540°$ (the pentagon angle sum), find the common difference.

$S_5 = \frac{5(2a_1 + 4d)}{2} = 540$: $\frac{5(160 + 4d)}{2} = 540 \Rightarrow 5(160 + 4d) = 1080 \Rightarrow 160 + 4d = 216 \Rightarrow 4d = 56 \Rightarrow d = 14°$.

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13 [Intermediate] • Target: Finding a term given the sum

The sum of the first n terms of an arithmetic sequence with $a_1 = 3$ and $d = 4$ is $S_n = 210$. Find n.

$S_n = \frac{n(2(3)+(n-1)4)}{2} = \frac{n(4n+2)}{2} = n(2n+1) = 210 \Rightarrow 2n^2 + n - 210 = 0$. Quadratic formula: $n = \frac{-1 \pm \sqrt{1+1680}}{4} = \frac{-1 \pm 41}{4}$. Taking the positive root: $n = 10$.

14 [Intermediate] • Target: Arithmetic sequences with negative common difference

A sequence begins $50, 44, 38, 32, \ldots$ Find the first negative term.

$d = -6$: $a_n = 50 - 6(n-1) = 56 - 6n$. Need $56 - 6n < 0 \Rightarrow n > 9.33$: first negative term is $a_{10} = 56 - 60 = -4$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Proving an arithmetic sequence property • Cross-Lesson: 4.02 Ellipses

Prove that for any three consecutive terms a_{n-1}, a_n, a_{n+1} of an arithmetic sequence, the middle term equals the average of its neighbors: $a_n = \frac{a_{n-1} + a_{n+1}}{2}$.

By definition, $a_{n-1} = a_n - d$ and $a_{n+1} = a_n + d$. So $\frac{a_{n-1} + a_{n+1}}{2} = \frac{(a_n - d) + (a_n + d)}{2} = \frac{2a_n}{2} = a_n$, confirming the middle term is the average of its neighbors.

16 [Challenging] • Target: Finding a term given the sum • Cross-Lesson: 4.03 Hyperbolas

In an arithmetic sequence, the 4th term is 17 and the 10th term is 47. Find the first term a_1, the common difference d, and the sum of the first 25 terms S_{25}.

Using $a_n = a_1 + (n-1)d$: $a_4 = a_1 + 3d = 17$, $a_{10} = a_1 + 9d = 47$. Subtract the first equation from the second: $6d = 30 \Rightarrow d = 5$. Substitute $d = 5$ into the first equation: $a_1 + 3(5) = 17 \Rightarrow a_1 = 17 - 15 = 2$. Find the sum of the first 25 terms using $S_n = \frac{n}{2}[2a_1 + (n-1)d]$:

$S_{25} = \frac{25}{2}[2(2) + 24(5)] = \frac{25}{2}[4 + 120] = \frac{25}{2}(124) = 25 \times 62 = 1550$.

17 [Challenging] • Target: Arithmetic sequences with negative common difference • Cross-Lesson: 5.01 Vectors: Components and Operations

The sum of the first n terms of an arithmetic sequence is given by $S_n = 2n^2 + 3n$. Find the 10th term a_{10}.

Step 1: Use relationship between S_n and a_n: $a_n = S_n - S_{n-1}$ Step 2: Method 1 (Direct evaluation): $S_{10} = 2(10^2) + 3(10) = 2(100) + 30 = 230$ $S_9 = 2(9^2) + 3(9) = 2(81) + 27 = 162 + 27 = 189$ $a_{10} = S_{10} - S_9 = 230 - 189 = \mathbf{41}$. Method 2 (General nth term formula):

$a_n = (2n^2 + 3n) - [2(n-1)^2 + 3(n-1)] = (2n^2 + 3n) - (2n^2 - 4n + 2 + 3n - 3) = 4n + 1$ For $n = 10$: $a_{10} = 4(10) + 1 = \mathbf{41}$.

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18 [Challenging] • Target: Proving an arithmetic sequence property • Cross-Lesson: 5.02 The Dot Product

In an arithmetic progression, the 3rd term is 14 and the 7th term is 34. (a) Find the first term a and the common difference d. (b) Find the sum of the first 25 terms.

(a) Use $u_n = a + (n-1)d$: $u_3 = a + 2d = 14$ $u_7 = a + 6d = 34$ Subtract the first equation from the second: $4d = 20 \Rightarrow d = 5$. Substitute back: $a + 2(5) = 14 \Rightarrow a = 4$. (b) Use the sum formula $S_n = \frac{n}{2}(2a + (n-1)d)$ with $n = 25$: $S_{25} = \frac{25}{2}(2(4) + 24(5)) = \frac{25}{2}(8 + 120) = \frac{25}{2}(128) = 25 \times 64 = 1600$

19 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 5.03 Parametric Equations

The sum of an arithmetic sequence is 150, with first term $a_1 = 5$ and last term $a_n = 25$. How many terms are in this sequence, and what is the common difference d?

Step 1 (Find n): Use the arithmetic series sum formula: $S_n = \frac{n}{2}(a_1 + a_n) \Rightarrow 150 = \frac{n}{2}(5 + 25) = 15n \Rightarrow n = \frac{150}{15} = \mathbf{10}$ Step 2 (Find d): Use the n-th term formula with $n = 10$: $a_{10} = a_1 + 9d \Rightarrow 25 = 5 + 9d \Rightarrow 9d = 20 \Rightarrow d = \mathbf{\frac{20}{9}}$

20 [Challenging] • Target: Arithmetic series • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Find the 12th term and the sum of the first 25 terms of the arithmetic sequence with $a_1 = -7$ and common difference $d = 4$.

Step 1: General term formula $a_n = a_1 + (n-1)d$: $a_{12} = -7 + (12-1)(4) = -7 + 44 = \mathbf{37}$. Step 2: Arithmetic series sum $S_n = \frac{n}{2}(2a_1 + (n-1)d)$: $S_{25} = \frac{25}{2}(2(-7) + 24(4)) = \frac{25}{2}(-14 + 96) = \frac{25}{2}(82) = 25 \times 41 = \mathbf{1025}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Arithmetic sequences

An arithmetic sequence has first term $a = 1$ and common difference $d = -4$. Find the sum of the first 3 terms.

a. -10

b. -7

c. -13

d. **-9 ✓**

$S_n = \frac{n}{2}(2a + (n-1)d) \Rightarrow S_3 = \frac{3}{2}(2 - 8) = -9$.

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22 [Foundational] • Target: Arithmetic series

An arithmetic sequence has a first term of $a_1 = 5$ and a common difference of $d = 3$. What is the value of the 6th term, a_6?

a. 20 ✓

b. 23

c. 17

d. 18

The general term of an arithmetic sequence is given by the formula: $a_n = a_1 + (n-1)d$ Substitute $a_1 = 5$, $d = 3$, and $n = 6$: $a_6 = 5 + (6-1) \cdot 3 = 5 + 5 \cdot 3 = 5 + 15 = 20$ Misconception Analysis: - 23 comes from using $a_1 + n \cdot d = 5 + 6(3) = 23$, forgetting to subtract 1 from n. - 17 is a_5, resulting from an off-by-one counting error. - 18 comes from directly multiplying $n \cdot d = 6 \cdot 3 = 18$, neglecting the initial term.

23 [Foundational] • Target: Arithmetic means and sequence properties

Consider the arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the fifth term of this sequence?

a. 17 ✓

b. 14

c. 20

d. 23

The formula for the n-th term of an arithmetic sequence is given by $a_n = a_1 + (n-1)d$, where a_1 is the first term and d is the common difference. In this case, $a_1 = 5$ and $d = 3$. We want to find the fifth term, so $n = 5$. Substituting these values into the formula: $a_5 = 5 + (5-1) \times 3$ $a_5 = 5 + (4) \times 3$ $a_5 = 5 + 12$ $a_5 = 17$

24 [Foundational] • Target: Applications

Find the sum of the first 8 terms of an arithmetic sequence with first term 5 and common difference 6.

a. 208 ✓

b. 202

c. 204

d. 206

$S_8 = \frac{n}{2}(2a + (n-1)d) = \frac{8}{2}(2(5) + (8-1)(6)) = 208$

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25 [Foundational] • Target: Visualizing an arithmetic sequence

An arithmetic sequence has first term $a_1 = -7$ and common difference $d = -1$. Find a_6.

a. -2

b. -12 ✓

c. 12

d. -13

$a_6 = a_1 + (6-1)d = -7 + 5(-1) = -12$

26 [Foundational] • Target: Arithmetic sequences in geometry

The first term of a geometric sequence is a. The common ratio is r. The sum of the first 3 terms is 21, and the sum of the first 6 terms is 189. If $r > 0$, find the value of a and r.

a. $a = 3, r = 2$ ✓

b. $a = 7, r = 1/2$

c. $a = 3, r = \sqrt{2}$

d. $a = 21, r = 1$

Let the geometric sequence be denoted by a_n, where $a_1 = a$ and the common ratio is r. The sum of the first n terms of a geometric sequence is given by $S_n = \frac{a(r^n - 1)}{r - 1}$ (for $req1$). We are given that $S_3 = 21$ and $S_6 = 189$. So, we have the equations: 1) $\frac{a(r^3 - 1)}{r - 1} = 21$ 2) $\frac{a(r^6 - 1)}{r - 1} = 189$ We can rewrite $r^6 - 1$ using the difference of squares factorization: $r^6 - 1 = (r^3)^2 - 1^2 = (r^3 - 1)(r^3 + 1)$. Substitute this into the second equation.

27 [Foundational] • Target: Finding a term given the sum

A geometric sequence has its first term a_1 and common ratio r. The sum of the first 3 terms is 21, and the sum of the first 6 terms is 189. Find the value of a_1 given that $req1$.

a. 1

b. **3** ✓

c. 7

d. 9

Let the geometric sequence be $a_1, a_1r, a_1r^2, \ldots$. The sum of the first n terms of a geometric sequence is given by $S_n = \frac{a_1(1 - r^n)}{1 - r}$ (for $req1$). We are given that the sum of the first 3 terms is 21: $S_3 = \frac{a_1(1 - r^3)}{1 - r} = 21$ (Equation 1) We are also given that the sum of the first 6 terms is 189: $S_6 = \frac{a_1(1 - r^6)}{1 - r} = 189$ (Equation 2) We can rewrite Equation 2 using the difference of squares factorization, $1 - r^6 = (1 - r^3)(1 + r^3)$.

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28 [Foundational] • Target: Arithmetic sequences with negative common difference

An arithmetic sequence has its n-th term given by $a_n = 3n + k$, where k is a constant. If the sum of the first 5 terms of this sequence is 65, find the value of k.

a. 1

b. **2** ✓

c. 3

d. 4

The sum of the first n terms of an arithmetic sequence is given by $S_n = \frac{n}{2}(2a_1 + (n-1)d)$ or $S_n = \frac{n}{2}(a_1 + a_n)$. In this case, $a_n = 3n + k$. So, $a_1 = 3(1) + k = 3 + k$. The common difference is $d = a_{n+1} - a_n = (3(n+1) + k) - (3n + k) = 3n + 3 + k - 3n - k = 3$. The sum of the first 5 terms is $S_5 = 65$. Using the formula $S_n = \frac{n}{2}(a_1 + a_n)$, we have $S_5 = \frac{5}{2}(a_1 + a_5)$. We need $a_5 = 3(5) + k = 15 + k$. Therefore, $65 = \frac{5}{2}((3 + k) + (15 + k))$. Multiply both sides b.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Proving an arithmetic sequence property

An arithmetic sequence has first term $a = 7$ and common difference $d = 3$. Find the 7th term.

a. 21

b. **25** ✓

c. 24

d. 26

Applying Series & the Binomial Theorem principles, $u_7 = a + (n-1)d = 7 + (7-1)(3) = 25$ Therefore, the verified correct option is 25.

30 [Intermediate] • Target: Arithmetic sequences

The 4th term of an arithmetic sequence is 19, and the common difference is 3. Find the first term.

a. 8

b. 9

c. 15

d. **10** ✓

Applying Series & the Binomial Theorem principles, $a = u_n - (n-1)d = 19 - (4-1)(3) = 10$ Therefore, the verified correct option is 10.

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31 [Intermediate] • Target: Arithmetic series

An arithmetic sequence has first term 10 and last term 40 with 7 terms. Find the sum.

a. 140

b. 175 ✓

c. 280

d. 350

Applying Sequences & Series principles, $S_7 = 7(10 + 40)/2 = 175$. Therefore, the verified correct option is 175.

32 [Intermediate] • Target: Arithmetic means and sequence properties

What is the common difference of the arithmetic sequence $0, -2, -4, -6, \ldots$?

a. 0

b. -2 ✓

c. -4

d. -3

Applying Sequences & Series principles, $d = -2 - 0 = -2$ Therefore, the verified correct option is -2.

33 [Intermediate] • Target: Applications

An arithmetic sequence has first term 6 and common difference 5. Find n such that $S_n = 1403$.

a. 19

b. 25

c. 27

d. **23** ✓

Using $S_n = \frac{n}{2}(2a + (n-1)d)$ with $S_{23} = 1403$, solving gives $n = 23$.

34 [Intermediate] • Target: Visualizing an arithmetic sequence

A geometric sequence has first term $a = 1$ and common ratio $r = 2$. Find the sum of the first 5 terms.

a. **31** ✓

b. -31

c. 32

d. 16

$S_n = \frac{a(r^n - 1)}{r - 1} \Rightarrow S_5 = \frac{((2)^5 - 1)}{2 - 1} = 31.$

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35 [Intermediate] • Target: Arithmetic sequences in geometry

In an arithmetic sequence, $a_4 = -1$ and $a_8 = -9$. Find the common difference d.

a. **-2 ✓**

b. -3

c. -6

d. 6

$d = \frac{a_8 - a_4}{8 - 4} = \frac{-9 + 1}{4} = -2.$

36 [Intermediate] • Target: Finding a term given the sum

The 12th term of an arithmetic sequence is 58, and the first term is 14. Find the common difference.

a. **4 ✓**

b. 1

c. 6

d. -1

$d = \frac{u_n - a}{n - 1} = \frac{58 - (14)}{12 - 1} = 4$

37 [Intermediate] • Target: Arithmetic sequences with negative common difference

Consider an infinite geometric series with first term a and common ratio r, where $|r| < 1$: $a + ar + ar^2 + ar^3 + \ldots = 12$ If each term in the series is squared, the sum of the resulting infinite geometric series is: $a^2 + a^2r^2 + a^2r^4 + a^2r^6 + \ldots = 72$ What is the value of the common ratio, r?

a. $\frac{1}{3}$ **✓**

b. $\frac{1}{2}$

c. $\frac{2}{3}$

d. $\frac{1}{4}$

Step 1: Use the sum formula for an infinite geometric series with $|r| < 1$: $S = \frac{a}{1 - r} = 12 \Rightarrow a = 12(1 - r)$ Step 2: When each term is squared, the new series has first term a^2 and common ratio r^2. Since $|r| < 1$, we also have $r^2 < 1$. Its sum is: $S_{\text{squared}} = \frac{a^2}{1 - r^2} = 72$ Step 3: Factor the denominator using the difference of squares, $1 - r^2 = (1 - r)(1 + r)$: $\frac{a^2}{(1 - r)(1 + r)} = (\frac{a}{1 - r})(\frac{a}{1 + r}) = 72$ Substi.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Proving an arithmetic sequence property • Cross-Lesson: 5.02 The Dot Product

In a geometric sequence of positive real numbers, the sum of the first two terms is 18, and the sum of the first four terms is 90. What is the fifth term of the sequence?

a. 48

b. 72

c. 96 ✓

d. 192

Let a_1 be the first term and r be the common ratio of the geometric sequence. Since all terms are positive real numbers, $a_1 > 0$ and $r > 0$. The sum of the first two terms is: $S_2 = a_1 + a_1 r = a_1(1 + r) = 18$ The sum of the first four terms is: $S_4 = a_1 + a_1 r + a_1 r^2 + a_1 r^3 = a_1(1 + r)(1 + r^2) = 90$ Substituting $a_1(1 + r) = 18$ into the equation for S_4: $18(1 + r^2) = 90$ $1 + r^2 = 5 \Rightarrow r^2 = 4$ Since $r > 0$, we have $r = 2$. Now substitute $r = 2$ back to solve for a_1.

39 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 5.03 Parametric Equations

An arithmetic sequence begins with the terms: $8, 13, 18, 23, \ldots$ What is the 15th term of the sequence?

a. 78 ✓

b. 83

c. 73

d. 117

To find the 15th term of an arithmetic sequence, use the explicit formula for the nth term: $a_n = a_1 + (n - 1)d$ 1. Identify the first term: $a_1 = 8$. 2. Find the common difference d by subtracting consecutive terms: $d = 13 - 8 = 5$ 3. Substitute $n = 15$, $a_1 = 8$, and $d = 5$ into the formula: $a_{15} = 8 + (15 - 1)(5) = 8 + 14(5) = 8 + 70 = 78$ Common Misconceptions: - Choice B (83) results from forgetting to subtract 1 from n, computing $8 + 15(5) = 83$. - Choice C (73) results from off-by-one in.

40 [Challenging] • Target: Arithmetic series • Cross-Lesson: 6.01 Polar Coordinates and Conversions

A geometric sequence has its first term $a_1 = 3$ and a common ratio $r = 2$. What is the third term, a_3, of this sequence?

a. 6

b. 9

c. **12** ✓

d. 15

The formula for the n-th term of a geometric sequence is $a_n = a_1 \cdot r^{n-1}$. Given $a_1 = 3$ and $r = 2$, we want to find the third term, a_3. Substituting $n = 3$ into the formula: $a_3 = a_1 \cdot r^{3-1} = a_1 \cdot r^2$ $a_3 = 3 \cdot 2^2$ $a_3 = 3 \cdot 4$ $a_3 = 12$. Therefore, the third term is 12.

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41 [Challenging] • Target: Arithmetic means and sequence properties • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the first term and common difference of the AP whose pth term is $3p + 2$.

a. $a_1 = 2, d = 3$

b. $a_1 = 5, d = 3$ ✓

c. $a_1 = 5, d = 2$

d. $a_1 = 3, d = 2$

Applying Sequences & Series principles, $a_p = 3p + 2 \Rightarrow a_1 = 5, d = 3$. Therefore, the verified correct option is $a_1 = 5, d = 3$.

42 [Challenging] • Target: Applications • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

The 5th term of the arithmetic sequence $2, 5, 8, 11, \ldots$ is:

a. 15

b. 13

c. 12

d. 14 ✓

Applying Sequences & Series principles, $a_n = 2 + (n - 1)3$. $a_5 = 2 + 12 = 14$. Therefore, the verified correct option is 14.

43 [Challenging] • Target: Visualizing an arithmetic sequence • Cross-Lesson: 7.02 Logarithms and Their Properties

An arithmetic sequence has $a_3 = 0$ and $a_5 = -4$. Find S_{12}, the sum of the first 12 terms.

a. -92

b. **-84** ✓

c. -85

d. -79

$d = \frac{-4-0}{2} = -2$, $a_1 = 4$. $a_{12} = 4 + (12 - 1)(-2) = -18$. $S_{12} = \frac{12(4-18)}{2} = -84$.

44 [Challenging] • Target: Arithmetic sequences in geometry • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

An arithmetic sequence has $a_1 = 3$ and $d = 6$. If the sum of the first n terms is 300, find n.

a. 11

b. 9

c. 5

d. 10 ✓

$S_n = \frac{n}{2}(2a_1 + (n - 1)d) = 300$, and solving the quadratic gives $n = 10$.

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45 [Challenging] • Target: Finding a term given the sum • Cross-Lesson: 7.04 Applications: Growth and Decay

Find the 15th term of an arithmetic sequence where $a_3 = 11$ and $a_7 = 23$.

a. 39

b. 51

c. 47 ✓

d. 43

$d = (23 - 11)/4 = 3$. $a_3 = a_1 + 2d \Rightarrow a_1 = 5$. $a_{15} = 5 + 14 \cdot 3 = 47$.

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